98 recall that electrons are released in the process of thermionic emission and explain how they can be accelerated by electric and magnetic fields.	

10.5 Probing the Heart of Matter (PRO)

Probing the Heart of Matter (PRO)

An area of fundamental physics that is the subject of current research, involving the acceleration and detection of high-energy particles and the interpretation of experiments:

- alpha scattering and the nuclear model of the atom
- accelerating particles to high energies
- detecting and interpreting interactions between particles
- the quark-lepton model.

Students study the development of the nuclear model and the quark-lepton model to describe the behaviour of matter on a subatomic scale.

There are opportunities to develop ICT skills using the internet and computer simulations.

Students learn how modern particle physics research is organised and funded, and hence have opportunities to consider ethical and other issues relating to its operation.

Students will be assessed on their ability to:	Suggested experiments
76 derive and use the expression $E_k = p^2/2m$ for the kinetic energy of a non-relativistic particle	
77 analyse and interpret data to calculate the momentum of (non-relativistic) particles and apply the principle of conservation of linear momentum to problems in one and two dimensions	
79 express angular displacement in radians and in degrees, and convert between those units	
80 explain the concept of angular velocity, and recognise and use the relationships $v = \omega r$ and $T = 2\pi/\omega$	

Students will be assessed on their ability to:	Suggested experiments
81 explain that a resultant force (centripetal force) is required to produce and maintain circular motion	
82 use the expression for centripetal force $F = ma = mv^2/r$ and hence derive and use the expressions for centripetal acceleration $a = v^2/r$ and $a = r\omega^2$	Investigate the effect of m, v and r of orbit on centripetal force
85 use the expression $F = kQ_1Q_2/r^2$, where $k = 1/4\pi\varepsilon_0$ and derive and use the expression $E = kQ/r^2$ for the electric field due to a point charge	Use electronic balance to measure the force between two charges
99 explain the role of electric and magnetic fields in particle accelerators (linac and cyclotron) and detectors (general principles of ionisation and deflection only)	
100 recognise and use the expression $r = p/BQ$ for a charged particle in a magnetic field	
101 recall and use the fact that charge, energy and momentum are always conserved in interactions between particles and hence interpret records of particle tracks	
102 explain why high energies are required to break particles into their constituents and to see fine structure	
103 recognise and use the expression $\Delta E = c^2\Delta m$ in situations involving the creation and annihilation of matter and antimatter particles	
104 use the non-SI units MeV and GeV (energy) and MeV/c^2, GeV/c^2 (mass) and atomic mass unit u, and convert between these and SI units	

Students will be assessed on their ability to:	Suggested experiments
105 be aware of relativistic effects and that these need to be taken into account at speeds near that of light (use of relativistic equations not required)	
96 use the terms nucleon number (mass number) and proton number (atomic number)	
97 describe how large-angle alpha particle scattering gives evidence for a nuclear atom	
107 write and interpret equations using standard nuclear notation and standard particle symbols (eg π^+, e^-)	
106 recall that in the standard quark-lepton model each particle has a corresponding antiparticle, that baryons (eg neutrons and protons) are made from three quarks, and mesons (eg pions) from a quark and an antiquark, and that the symmetry of the model predicted the top and bottom quark	
108 use de Broglie's wave equation $\lambda = h/p$	